Product attributes and the model of emotional design: How the product development engineers perceive product features?

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Abstract

In design practice emotions elicited by product appearance are often considered to be intangible and, therefore, quite hard to manipulate, even less to treat in a level of discussions. This arises from the fact that the emotions are something that a person has, rather than a social fact. Another usually stated emotional-related issue is that emotions are very difficult to define. This leads to define emotions as a subjective state of mind that a person is ambiguously aware of. However, it would be advantageous to assume that actors participating in a product development process would be able to communicate emotions linked to the products appearance using the 'signals' based on product attributes. The most common way of signalling is through spoken language and different attributes can be used to define the technical, aesthetical and other perceived aspects of the products. This is in line with Donald Norman's (2004) well known model of emotional design - the ways people perceive emotional design in products. This happens principally in three levels: visceral, behavioural and reflective.

This paper presents the empirical findings on how product development engineers link the pre-defined and emotionally orientated product attributes to products. The questions studied are how the three levels of emotional design can be linked to attributes in different products. The study is based on visual data collected from the annual Finnish furniture and interior decoration fair 'Habitare', in 2004. The study is based on 90 photographs taken by product development engineers working in R & D unit at the global manufacturing company. The company is specialised in manufacturing transportation systems. The data embodies 90 images, and it includes also short participant interviews conducted after collecting the visual data. The test persons, altogether 15 product development engineers in 5 groups, participated in the study. The product development engineers were to identify 20 given attributes based on the classification of Johnson et al. (2003) by finding products that represent those attributes.

The result of the study shows that the groups had tendency to link attributes to the products on behavioural level rather than on the reflective level. Applying the slightly revised Norman's model, the product's selection criteria were clearer in visceral and behavioural levels than in reflective level.

Based on this study, designers may find it worthwhile to include emotions in the intentions of design efforts. In design process this can be concretely utilised by using certain product attributes that will probably induce desired responses. On the other hand, emotional responses may incite consumers or customers to select a particular product based on visceral, behavioural and reflective emotions. Therefore emotions can be seen having a considerable influence on purchase decisions, both in consumer and investment goods businesses. From an organisational point of view, the emotions should be taking a firmer place in a design process. Based on this study, to designers is offered a classification of product attributes that should aid in discussing the abstract concept of 'emotional concerns' and how design can be seen through different product attributes. Understanding the particular product attributes on abstract level, designing a product that 'feels good', 'has desired feeling' or to design a product that 'follows corporate brand values' is made visible in product development organisation. It can be seen as a consequence of a development path in which more and more designers are challenging consumers and customers to discover the emotional impact in their lives.

Key words: design features, emotions, emotional design, product attributes, product evaluation

1. Introduction

It is commonly acknowledged that emotions are in many ways immanent in everyday life. For most of us emotions guide, enrich and even provide the meaning to everyday existence (Cacioppo et al. 2000, 173; Diener & Lucas 2000). However, emotions can also be seen relevant for a product experience. This takes place because emotions imply to a one-to-one relationship between the affective state and a particular product or object (see Frijda 1986). It is also noticed that the other affective states, such as feelings and moods, do not particularly involve a specific object (Desmet and Hekkert 2002). This is the reason emotions are an interesting topic to study in the context of design. However, when emotions are brought under more precise study several unanswered issues are emerging.

Especially in design emotions elicited by product appearance are often considered to be intangible and therefore quite hard to manipulate, much less to approach as a level of discussions. This arises from the fact that the emotions are something that person rather has as a social fact. This leads to define emotions as a subjective state of mind taking place in various contexts. However, it would be advantageous that actors participating product development process would be able to communicate about emotions linked to the products appearance at least on some level.

The questions studied in this paper are closely linked to Norman's (2004) concept of emotional design. The main question is how the different levels of emotional design suggested by Norman are linked to the given product attributes? In addition, an interesting question is whether the interpretations of the attributes' differ among rather homogenous group of people studied in this paper.

However, before one can speak about emotions, one must be able to characterise and define them. Unfortunately, this is a problem that belongs to those yet unsolved although affect, emotion, and mood all play a powerful role in human cognition (Norman 1992; 1998). While the concept of emotion appears to be generally understood, it is quite difficult to come up with a definition that covers phenomenon acceptably. It is difficult to see why products could not elicit all kinds of emotions. Emotions are elicited not only by the product's aesthetics, but also by other aspects, such as the product's function, associated product attributes, preceding

behaviour, and various contexts and meanings. Emotions are always experiences that individual undergoes in different situations while it is possible to have several, also contradictory, emotions towards the same product. (See Desmet 2002; Desmet 2003a; Desmet 2003b; Desmet et al. 2004). As there seems to be no empirical solution to this debate of defining emotions, the most favoured solution, is to say that emotions are best treated as a multifaceted phenomenon consisting of components of behavioural reactions, expressive reactions, physiological reactions, and subjective feelings (Desmet 2003a, 111-112).

Desmet et al. (2004) suggest that people differ with respect to their emotional responses towards a given product. One person may be inspired by the product, whereas the same product may disappoint another person, or at least the responses invoked are different from person to person. This may be the generally accepted 'truth', but it can be also stated that product at hand may have some distinct emotional attributes that a group of people seem to hold the same (Desmet et al. 2004). This paper investigates basically this phenomenon.

The purpose of this study is to aid the process of product design. Often, the product requirements are specified mainly through technical attributes because emotional attributes are considered too intangible to be able to guide product development. Furthermore, the emotional attributes have no measurable value. The goal of this study is to discuss the possibilities to express emotions linked to the product through attributes. This means, at least, that "*I like*..." and "*I don't like*…" kind of arguments could be left behind and develop a new way of argumentation regarding product appearance in the product development teams.

Above mentioned setting provides a situation for better mutual interpretation of product attributes and possibility to evaluate different interpretations the attributes evoke. On a more general level the purpose is to apply and evaluate Donald Norman's (2004) model of emotional design to a case where product attributes are given and emotional perception of products is taken as a premise for product evaluation. Norman's model provides methodological assistance in sorting out and grouping the research data. The classification provided by this study can also aid designers to some extent to suggest emotions they want to arouse with their designs.

2. Research method, materials and procedure

In this paper the interpretation theory is a slightly revised application of Donald Norman's emotional design model. Norman's model is divided roughly to three different levels that emotional design can have effects on. Those levels are: visceral, behavioural, and reflective.

The first, *the visceral level of design,* is about how things look and feel. Visceral appeal is fast, sometimes instant, and most products have it to some degree. (Norman 2004, 67). On the second, *the behavioural level of design*, the emotional impact is guided mostly by function and usability. In this level product appearance does not carry such a remarkable meaning as performance does. The third, *the reflective level of design,* is about the various meanings of things (Norman 2004, 70). The reflective level is about message and culture. It is also the stage in which things like brand image and marketing come into play. Products are sold not on their functionality but on things like prestige and exclusivity (Norman 2004, 83).

This study is a part of the Proactive Design research project (2002-2004) financed by the Finnish Funding Agency for Technology and Innovation, TEKES. In this project, the particular attention was paid to study how industrial design as a competence was included to company activities to improve global competitiveness. The research subjects were four global Finnish enterprises, leaders in their field, representing transportation systems, construction elements, and paper and timber industry.

The data studied in this paper has been collected in an Industrial Design Theme Day event organized for the staff at R & D department of a company specialised in manufacturing transportation systems. Researchers were asked to set up research-related activity for participants during their visit in the Habitare 2004 – furniture fair which was the part of the agenda for the Industrial Design Theme Day[1].

The company's project managers have earlier informed the researchers that it is difficult to discuss the proposals and drafts made by industrial design among the product development team members. Often the team just vote for proposals finding that practice only possible way to prioritise those proposals. The objective of the event was to improve the product

[1] Habitare is Finland's biggest event in the furniture and interior decoration field.

development engineers capability to exchange views regarding the product appearance. The other goal was to explore how the interpretations of attributes diverge (see Figure 1 and Appendix A).

Test arrangement

All together 15 engineers attended to the theme day at the Habitare -fair. The participants formed a quite homogenous group. They were all Finnish, lived in the same small town in Southern-Finland, had similar educational background, and have worked several years at the R & D unit of the same company. The task given to the participants was to track emotional responses elicited by products in order to practice to use attributes as signaling emotions. They worked in five groups, three people in each group, in order to assure the discussion about the relations between product and attributes. The groups had digital cameras and they were given 20 attributes based on a classification by Johnson et al. (2003; also Appendix A). The given attributes were chosen from Johnson's list based on the knowledge that they were used often in various product development documents in the company. The brand attributes of the company, *innovative*, *reliable* and *responsible,* were included on the list.

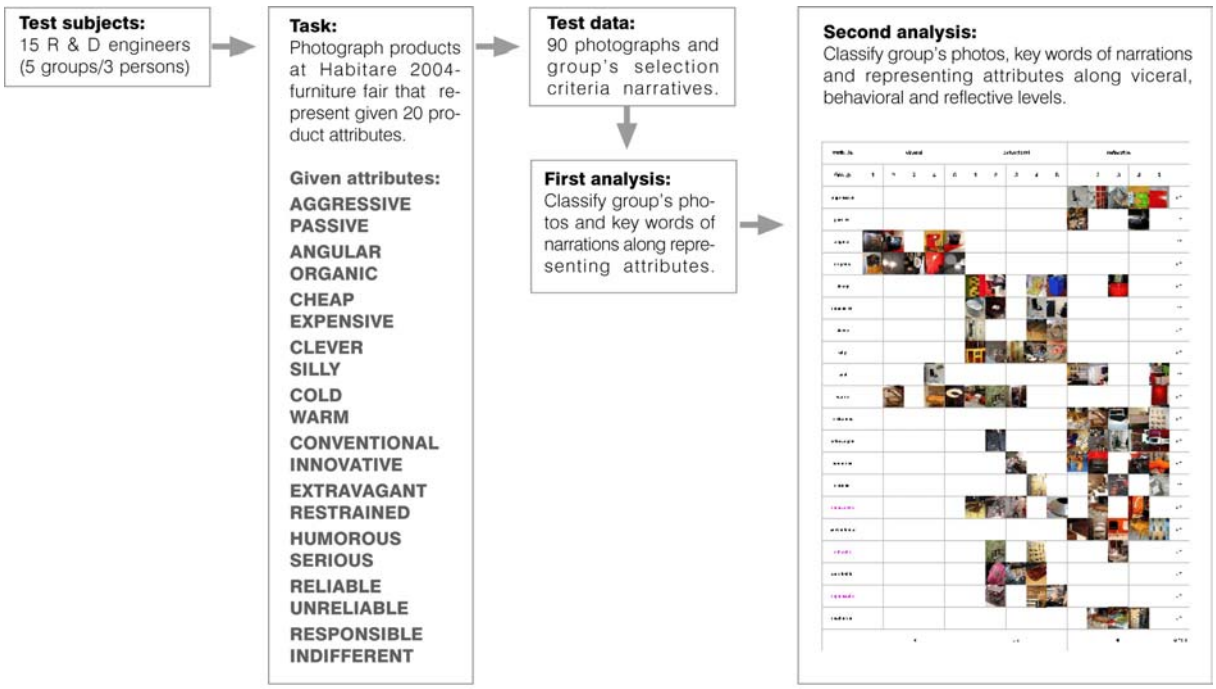


Figure 1. Test arrangements and classifying the data.

The collected data consists of 90 images photographed by the groups, and it includes short interviews of the groups conducted few hours after they collected the visual data. The aim of the interviews was to understand the group's criteria to link the certain attribute to the certain product. Some groups were not able to link any attribute to a particular product and some groups linked several products to one attribute. In latter case there are two or more images in the square of representing group's choice (see Figure 1 and Appendix A).

Classifying and analysing the data

First the data, product images and key words of selection criteria, was classified according to attributes (see Figure 1). Main finding from this classification was that some attributes were easier for the groups to link to the products. For example, all groups were able to identify such attributes as *organic*, *aggressive* or *innovative*. On the contrary, the attributes *clever*, *passive*, *reliable* or *responsible* were more problematic since all groups could not find any products signalling those attributes. Furthermore, the groups' selection criteria of products signalling particular attribute differ from each other or were similar without any clear logic. Johnson et al. (2003) proposed distribution to Aesthetics-Sensory and Perception-Symbolic attributes did not shed more light on this phenomenon.

However, it became clear that what was needed was a more specific system for classifying the data in order to generate more specific information. Also the group's selection criteria for choosing products have to be integrated to some theoretical model to reflect upon. As a ground of new classification we decided to utilise Donald Norman's (2004) well known theory of emotional design - the ways people perceive emotional design in products - described in detail in the beginning this chapter. However, classifying our data according to these three levels was somewhat problematic since Norman's categories are left open to various interpretations and they are not tied up very clearly to the solid examples or illustrations. Next chapter will provide more detailed discussion on how the boundaries were set in order to construct a more suitable model to the interpretation task at hand.

3. Emotions, product attributes and emotional design

The modifications made to the Norman's model are mostly on the level of specifications; the basic classification still remains the same. Based on the empirical data, the visceral level is embodying on easily recognisable features. These visual product attributes are mainly basic geometric forms or sense of touch that are perceived quite universally through various cultures. The visceral product attributes appear to be instantly recognisable and there is seldom a difference in the interpretations of test groups. The visceral level seems to be the first and easiest way to react to a product (see chapter 3.1).

The behavioural level is closest to Norman's original model. It emphasises practicalities of the product: for example usability, manufacturability and experiences of use. The product's function is the main factor linking interpretations on this level. If visceral level does not provide agreeable explanation, the test groups started to interpret attributes through the function of the product. Employing this level requires that the test persons have used the particular products or they have obtained information about product's functionality some other way (see chapter 3.2).

Furthermore, it seems that if the given attribute could not be explained on a visceral or behavioural level, test group had to interpret the attribute on a more abstract level. This is when the reflective interpretations take place. The reflective level is manifesting the beliefs and very subjective experiences or explanations. Usually the test groups took one feature of the product and reflected the attribute upon it in some way. There was no clear reason expressed which feature was chosen and why. On this level the attributes reflect the social and cultural ideas that may connect very special group of people. It is typical to the reflective level of product emotions that some "difficult" attributes are linked to the product with the most peculiar explanations given by test groups (see chapter 3.3).

Desmet and Hekkert (2002) found that there is often not a one-to-one relationship between the appearance of a product and the emotional responses it elicits. Emotions are not elicited by product characteristics as such but by construal based on these characteristics. By their definition, the construal is highly personal. (Desmet and Hekkert 2002). This is in line with the reflective level suggested in this paper. In reflective level socially and culturally determined factors are affecting to emotions, and these factors change from person to person.

However, this study is conducted to understand how single product attribute can be experienced and with what kind of a product this particular attribute is combined with.

Regardless, a few conditions are relevant to outline prior to describing the levels more thoroughly. Firstly, the emotional descriptions produced in this paper are ideal-types; only one attribute describes the product. Secondly, the attributes given in study setting are describing the whole product, in which case product is seen in quite an abstract manner. Thirdly, these descriptions can be socially constructed or culturally bound. For example, the products linked to the attribute *reliability* may have been chosen by different selection criteria in cultural comparison.

3.1 Visceral product attributes – aesthetic, simple and universal form?

During the first analysis it was noticed that the explanation for linking the products to the attributes *angular* and *organic* were same in every group. In other words, the groups' selection criteria did not vary. 'Organic' products were selected on account of the form being based on curves while all 'angular' products had strong edge angles. Besides that, none of the groups associated behavioural or reflective product interpretations to these attributes (see Figure 2).

The above metioned findings led us to link geometrical characters of the form to the visceral level. Accordingly, visceral level includes artefacts with physical, concrete attributes that are able to recognise. These attributes are defining products in a straightforward manner and do not leave room for negations.

attribute	viceral					behavioral					reflective					
Group	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	
angular																4/5
organic																5/5

Figure 2. Angular and organic products in the visceral level of design

However, the possibility to link an attribute to the physical facts of a product does not necessarely mean that it is the only option for interpretation. Collocation such as *warm-cold* is an example of that approach (see Figure 3). Both attributes can be linked to the material of products through the sense of touch as some groups did. Group 4 chose the products made of steel (washbasin and chair) that are cold to touch, while the rest of the groups linked 'cold' to the reflective level by choosing white painted, very simple (angular) and materially hard products to mirror the attribute 'cold'.

When observing the attribute 'warm' interesting issues emerge. Group 4 chose wooden product to represent the attribute 'warm' (same washbasin). This choice was consistent with their choice for attribute 'cold'. The group explained their choice: "*wood is warm to touch like steel is cold*". Group 5 has also consistent choice since attribute 'cold' was linked to the color (white), material (hard) and form (angular), the attribute 'warm' was red and soft object. Groups 1, 2 and 3 all selected the products that are used to produce heat such as a sauna, a tile stove and various shawls or blankets. They were all about the use and were classified to the behavioral level.

attribute	viceral					behavioral					reflective					
Group	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	
cold																4/5
warm				 		  										5/5

Figure 3. Warm and cold products in the visceral, behavioral and reflective levels of the design

This finding evokes the question why there were no behavioral level product interpretations regarding the attribute 'cold' with groups 1, 2 and 3, because the groups seemed to have an intention to keep their selections consistent regarding the collocations. The simple answer to this may be, that there were no cold producing domestic electrical appliances such as

refrigerators or fans being presented in the Habitare –fair. However, according to these two cases it seems that the groups attempted to link the attribute to the product through the mutually accepted physical characters on the visceral level. If they were not able to identify the products through an attribute located in the visceral level, they tried to link the attribute to the product that has an experience of use taking place on behavioral level. This finding is confirmed through next examples as well.

3.2 Behavioural product attributes – previous experiences, usage and usability?

As already mentioned in the previous chapters, behavioral level attributes are those that are possible to retrieve to pragmatic activities such as usability, manufacturability and experience of use. All but one group tied up the attributes *cheap-expensive* and *clever-silly* to the behavioral level (see Figure 4). Looking only at the images it is difficult to understand the logic behind the selection. However, the groups' explanations clarified their selection criteria. The selection criteria of the products regarding the attributes cheap and expensive were "*low and high manufacturing costs*". The products were either made out of expensive or cheap materials or manufacturing methods are difficult (expensive) or easy (cheap). For example, group 2 photographed the slice of bread and butter with ham symbolizing the attribute cheap in a reason that *"the ingredients cost hardly anything"* and office chair was chosen to illustrate the attribute expensive by group 4 because *"the production methods are so difficult that the chair will cost a fortune"*. Group 3 alone was linking the attribute 'cheap' to the reflective level by choosing, as they explained, the whirlpool bath because "*color made it look so 'cheap' that it is suitable to a brothel*".

The connotation clever-silly followed the same logic as cheap-expensive. The attribute clever was linked to the behavioral level by choosing a product which had construction solutions creating a new way to function or a new look to the old function. The artifacts with unnecessary decorations for use, such as ornament on the back of the chair, were considered representing the attribute 'silly'. Only the selection criteria of group 3 differ from others. In the image of group 3 was shower with small carpet that "*gets wet when using the shower, which is ridiculous*". Poor usability was considered 'silly' as well.

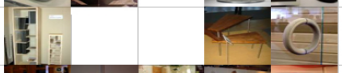
attribute	viceral					behavioral					reflective					
Group	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	
cheap																5/5
expensive																4/5
clever																3/5
silly																5/5

Figure 4. Cheap and expensive products in the behavioural and reflective levels of design. Clever and silly products in the behavioral level of design.

These examples show that if an attribute does not link the product on the visceral level and it is possible to connect to the behavioral one, the groups had no need for looking for the reflective level interpretation.

3.3 Reflective product attributes – culturally and socially meaningful interpretations?

In the reflective level the attributes manifest quite subjective perception of products. In this level the attributes reflect the social and cultural ideas that can also be connected to a special group of people as well.

The groups' selection criteria seem to settle to the reflective level if there was no a simple straightforward physical character or pragmatic activity available in order to link the attribute and the product to the previous levels. The attributes *aggressive* and *passive* are examples of the reflective level. There were several product features that the groups identified as aggressive: bright colours (especially red), angular or complex forms, and materials that originate from the animals that are aggressive by nature (chair coated with panther fur, see Figure 5). *"The chair is aggressive because it is coated with panther fur and the panther is hostile animal"* explained the group 3 about their choice. The main finding is that even among very homogenous group of people the interpretations of attributes vary to a great

extent in this reflective level. However, the members of a group were able to decide through discussion which features represented, for example, an aggressive product.

The attribute '*passive*' emerged to be the most difficult to link to the products in the Habitare –fair. Only groups 1 and 4 were able to produce images representing the attribute 'passive'. Instead of a product both groups chose a human character as an example. During the interview the groups explained that finding the products signalling passive attribute was impossible task. Group 1 chose a doll and group 4 chose a silent woman sitting alone. The doll and the woman were selected as 'passive' because they did not communicate in any way. The selections were explained afterwards: "*the doll cannot talk or move*" and "*woman does not talk, neither has she moved*".

attribute	viceral					behavioral					reflective					
Group	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	
aggressive																5/5
passive																2/5

Figure 5. Aggressive and passive products in the reflective level of design

Those two images representing the attribute 'passive' can be located to the reflective level if communication point towards social requirements. As these examples point out, some attributes are quite hard to find from the products', some of them being impossible for the groups studied in this occasion. On the reflective level the positioning was complex even though the interviews of the groups guided the interpretations.

4. Discussion

This paper presented the empirical findings on how certain product development engineers linked the given attributes to the products. The questions studied were how three levels of emotional design can be expressed through attributes and how these attributes are interpreted through products.

Based on this study, the test groups connected the given 20 attributes to products on a visceral level when ever it was possible. Thus, the visceral level perception is strongly linked to the first impression the product elicits. If the instant interpretation on the visceral level did not provide a satisfactory match, the groups tended to make the connection on the behavioural level and only after that on the reflective level. This convention or a way to perceive the relations of attributes and products (mathematical, functional, manufacturability) might differ in situations where a person in test groups would have different background than engineering. Although the test groups were homogenous, the basis of connecting attributes and products from Habitare –fair changed radically from group to another on the reflective level. There was also minor deviation on the behavioural level.

Connecting the attributes to various products occurred on different emotional levels. The classification of the photographs is based on the interpretation that the majority of the test groups made. Despite of the test group interviews, some linkages between the product's selection criteria and attribute were not clear. This might be a result of the fact that connecting written product attributes and emotions products' elicit can be cryptic. Another interpretation might state that some attributes employed in this study were too difficult or even impossible to identify through product appearance. These were such as the attributes of *responsible* or *reliability* that are chosen to represent the brand of the company.

Using emotions as an axiomatic is quite difficult in the context of this study. Exclusively the 'research design' of this study comprises the given product attributes, a bunch of different products and emotional responses of the test person interpreted in three levels – all affecting to the ways product design can be perceived in terms of emotional design. As Desmet have pointed out, there is rarely a one-to-one relationship between the design of a product and the emotion it elicits. Emotions are not elicited by a product as such, but by the appraised significance of this product for our concerns (Desmet 2002, 124). This is the unavoidable case

also in this study. As mentioned in the beginning of the paper, the basic fact seems to be well-known. Many products induce strong but contradictory emotions in different people (Desmet 2002; Desmet et al. 2001). In this study, the result is that products and the attributes can be perceived in three emotional levels each embodying different means of interpretation. Based on the empirical findings, these three levels seem to be structured somewhat hierarchically.

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Appendix A. Collected data classified according visceral, behavioural and reflective levels of design.

attribute	viceral					behavioral					reflective					
group	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	
aggressive																5/5
passive																2/5
angular																4/5
organic																5/5
cheap																5/5
expensive																4/5
clever																3/5
silly																5/5
cold																4/5
warm																5/5
restrained																5/5
extravagant																5/5
humorous																5/5
serious																4/5
innovative																5/5
conventional																5/5
reliable																3/5
unreliable																3/5
reponsible																3/5
indifferent																3/5
	14					36					40					90/100